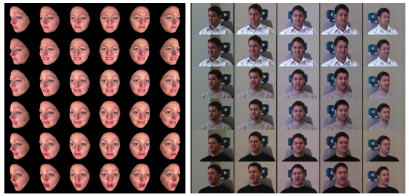
Pose-Invariant Facial Expression Recognition

Guang Liang, Shangfei Wang, and Can Wang
University of Science and Technology of China

FG 2021

Motivation



- Pose-invariant facial expression recognition is quite challenging due to variations in facial appearance and self-occlusion caused by head rotations.
- In this paper, we propose an adversarial multi-view subspace learning method for pose-robust facial expression recognition.

Main Idea

- Pose-specific feature extractors are trained to extract facial representations. Adversarial learning and similarity constraint are introduced to capture the correlations, and to minimize the distribution differences between different poses.
- The ensemble method is adopted to combine the predictions of multiple deep neural networks and the common expression classifier, so pose estimation is not required during testing.

Problem Statement

Training data:

- Let $D = \{x_i^1, x_i^2, \dots, x_i^T, y_i\}_{i=1}^N$ denote the training set, where $x_i^1, x_i^2, \dots, x_i^T$ represent the ith sample of different poses, T represents the number of poses, y_i represents the ground truth expression label, and N indicates the number of training samples. One training sample consists of face images from T poses with the same expression.
- Goal: learn a classifier $f = h \circ g$ to predict the expression label of an unseen sample with an arbitrary pose.

Proposed Method

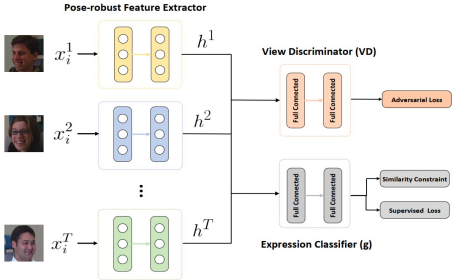


Fig.1 Framework of pose-invariant facial expression recognition.

◆ The framework includes a pose-robust feature extractor consisting of T deep networks, $h^1, h^2, \dots, h^T$, a view discriminator VD, and a common expression classifier g. We denote the parameters of the view-robust feature extractor as θ_h, the expression classifier g as θ_g and the view discriminator VD as θ_d.

- Classification loss
$$\mathscr{L}_{exp} = \sum_{t=1}^{T}\sum_{i=1}^{N} -y_i \log \hat{y}_i^t - (1-y_i)\log(1-\hat{y}_i^t)$$
- Similarity constraint
$$\mathscr{L}_{sim} = \sum_{p,q}^{T}\sum_{i=1}^{N} \left\| f^p(x_i^p, \theta_f) - f^q(x_i^q, \theta_f) \right\|^2$$
- Adversarial loss
$$\mathscr{L}_{adv} = \sum_{t=1}^{T}\sum_{i=1}^{N} -v_i \log(\mathbf{VD}(h_i^t(x_i^t, \theta_h^t), \theta_d)) - (1-v_i)\log(1-\mathbf{VD}(h_i^t(x_i^t, \theta_h^t), \theta_d))$$
- Overall loss
$$\mathscr{L}(\theta_h, \theta_g, \theta_d) = \mathscr{L}_{exp}(\theta_h, \theta_g) + \alpha \mathscr{L}_{sim}(\theta_h, \theta_g) - \beta \mathscr{L}_{adv}(\theta_h, \theta_d)$$

◆ To balance all pose-specific feature extractors, we aim to train the weights w of different networks from the training set. A training sample x_i with expression label y_i is used as the input of all feature extractors, while freezing the parameters of all the networks.

- Weight loss
$$\mathscr{L}_w = \sum_{i=1}^{N} -y_i \log \sum_{t=1}^{T} w_t \hat{y}_i^t - (1-y_i)\log(1-\sum_{t=1}^{T} w_t \hat{y}_i^t)$$
$$s.t. \sum_{t=1}^{T} w_t = 1, w_t \in (0,1)$$
- Predicted expression
$$\arg\max \frac{1}{T}\sum_{t=1}^{T} w_t g(h^t(x_i^{test}, \theta_h^t), \theta_g)$$

Experiments

- Databases: the Multi-PIE database, the BU-3DFE database, the SFEW database, and the AffectNet database.
- Experimental results and analyses

Method	Multi-PIE													
	0°	15°	30°	45°	60°	75°	90°	Avg.	−30°	−15°	0°	15°	30°	Avg.
Compare_{ms}	75.0	85.8	84.2	86.6	75.0	85.8	80.0	81.7	87.7	85.9	87.7	80.9	87.2	85.9
Compare_{sim}	75.0	86.6	87.5	83.3	80.8	86.6	84.1	83.4	89.1	87.7	89.1	84.5	91.3	88.3
Compare_{adv}	83.3	87.5	90.8	93.3	88.3	89.1	86.6	88.4	90.9	93.1	92.3	90.9	95.0	92.4
$\text{Ours}_{adv+sim}$	87.5	92.5	89.1	93.3	90.8	92.5	90.0	90.8	93.8	93.4	94.3	92.0	93.3	**93.3**
Method	BU-3DFE													
	−45°	−30°	−15°	0°	15°	30°	45°	Avg.	0°	30°	45°	60°	90°	Avg.
Compare_{ms}	79.5	82.7	81.6	81.0	83.3	82.9	81.0	81.8	72.3	76.4	73.3	75.6	78.7	75.3
Compare_{sim}	80.2	82.0	83.9	85.4	83.3	82.5	85.0	83.2	73.8	77.3	77.9	77.5	74.8	76.2
Compare_{adv}	88.1	87.2	86.7	87.5	88.5	86.7	87.5	87.5	82.1	81.2	80.8	82.9	79.8	81.4
$\text{Ours}_{adv+sim}$	89.9	89.5	90.3	91.4	89.4	88.6	87.2	89.7	80.8	81.8	85.4	84.5	81.4	**82.8**

- Comparison to related work

Method	Feature	Multi-PIE			BU-3DFE		
		Poses	Pose Pan	Acc	Poses	Pose Pan	Acc
Moore	LGBP	7	(0°,90°)	80.4	5	(0°,90°)	68.0
GSRRR	LBP^{u2}	7	(0°,90°)	81.7	5	(0°,90°)	66.0
KPSNM	HoG+LBP	7	(0°,90°)	83.1	5	(0°,90°)	78.8
LLCBL	Dense SIFT	7	(0°,90°)	86.3	5	(0°,90°)	74.6
DNND	SIFT	7	(0°,90°)	85.2	5	(0°,90°)	80.1
EPRL	DNN	7	(0°,90°)	87.1	5	(0°,90°)	73.1
IPFR	DNN	7	(0°,90°)	88.4	5	(0°,90°)	81.8
$\text{Ours}_{adv+sim}$	DNN	7	(0°,90°)	**90.8**	5	(0°,90°)	**82.8**
DS-GPLVM	Landmark	5	(−30°,30°)	90.6	–	–	–
HBTM	SIFT	5	(−30°,30°)	90.2	7	(−45°,45°)	79.1
JPEM	DNN	5	(−30°,30°)	91.8	7	(−45°,45°)	81.2
GGPI	DNN	5	(−30°,30°)	92.1	7	(−45°,45°)	82.0
IPFR	DNN	5	(−30°,30°)	92.6	7	(−45°,45°)	85.1
Xu	DNN	3	(−15°,15°)	93.1	–	–	–
$\text{Ours}_{adv+sim}$	DNN	5	(−30°,30°)	**93.3**	7	(−45°,45°)	**89.7**

Method	Train	Test	Emotions							Avg.
			AN	DI	FE	HA	NE	SA	SU	
DS-GPLVM	BU-3DFE	SFEW	25.9	28.2	17.2	43.0	14.0	33.3	11.0	24.7
JPEM	BU-3DFE	SFEW	30.9	22.0	19.6	50.9	19.2	28.0	15.5	26.6
$\text{Ours}_{adv+sim}$	BU-3DFE	SFEW	25.7	15.6	39.2	42.9	21.3	36.4	14.7	**28.1**
IPA2LT	AffectNet	SFEW	-	-	-	-	-	-	-	58.1
$\text{Ours}_{adv+sim}$	AffectNet	SFEW	69.2	30.4	32.3	91.8	68.3	52.5	70.8	**59.5**
IPA2LT	AffectNet	AffectNet	-	-	-	-	-	-	-	56.5
$\text{Ours}_{adv+sim}$	AffectNet	AffectNet	84.3	27.6	27.5	92.6	45.3	51.1	77.3	**57.8**

- Comparison of ensemble averaging method

Train	Test	Number of T		
		2	3	5
BU-3DFE	SFEW	21.2	26.3	28.1
AffectNet	SFEW	48.3	55.7	59.5
RAF-DB	ExpW	61.0	65.6	69.1
AffectNet	AffectNet	46.4	53.9	57.8

Conclusions

- We propose an adversarial multi-view subspace learning method for pose-robust facial expression recognition.
- Experimental results demonstrate our superiority.

References

[18] Peng et al. Moment matching for multi-source domain adaptation.
[24] Wang et al. Identity-and pose-robust facial expression recognition through adversarial feature learning.